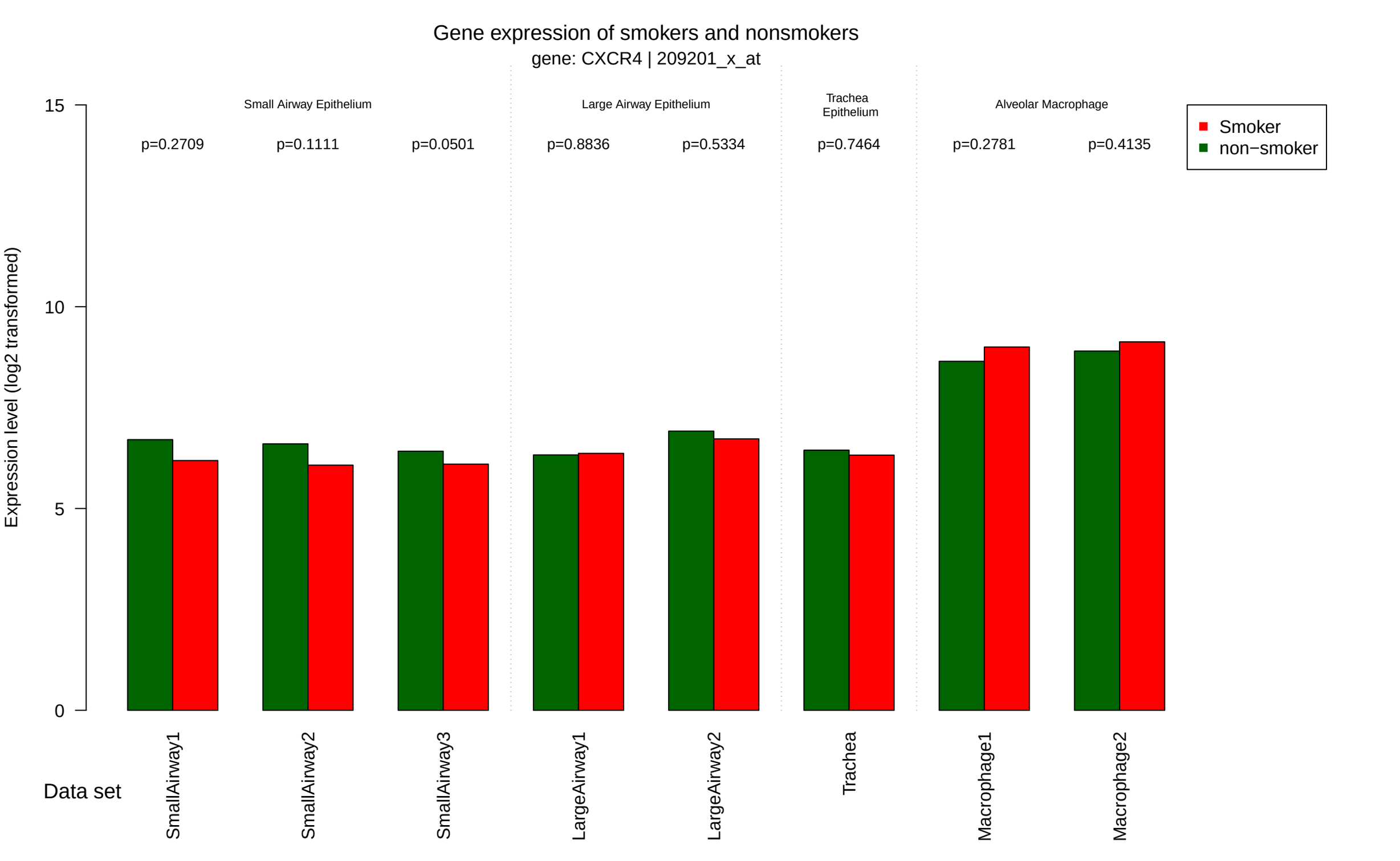
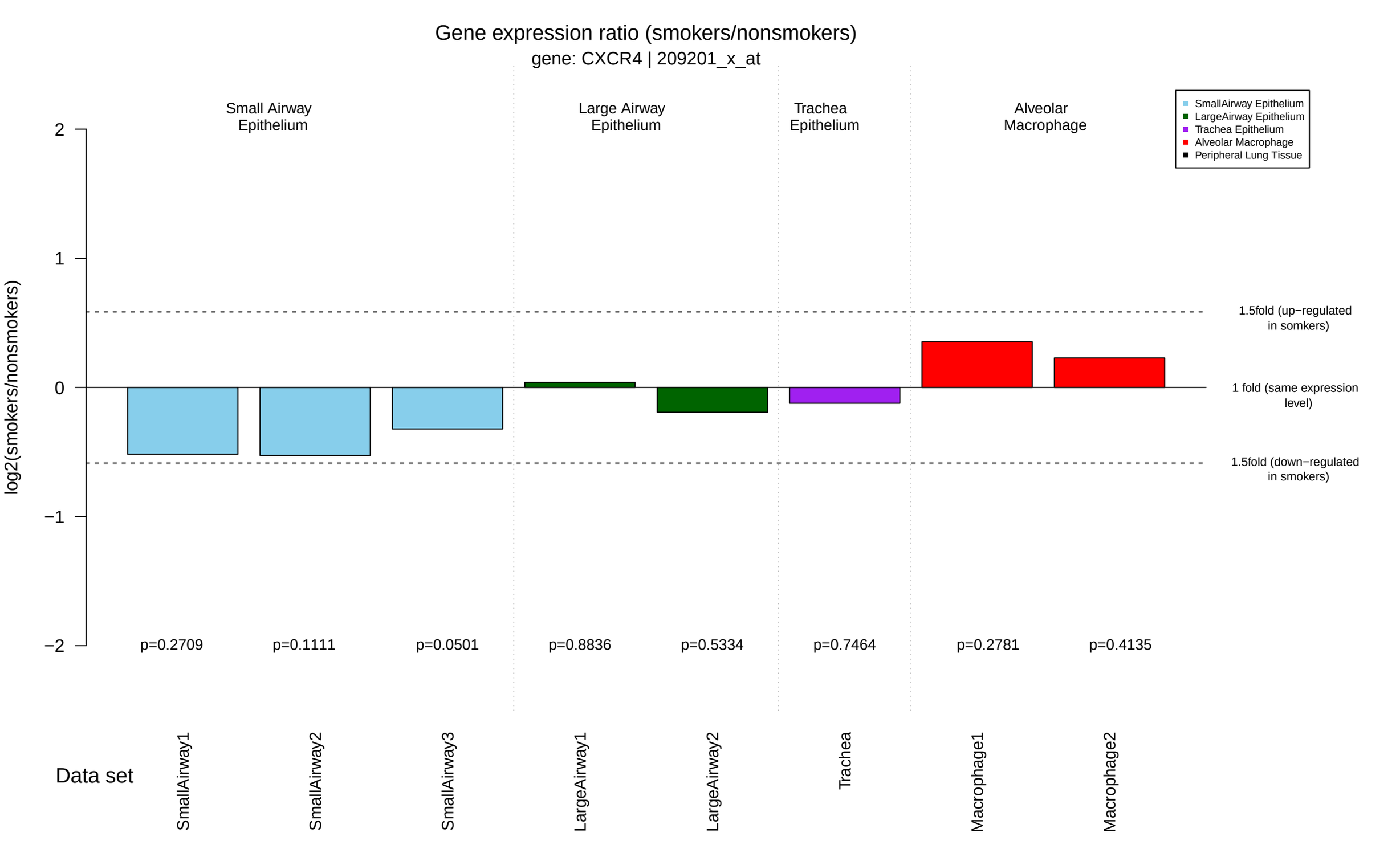
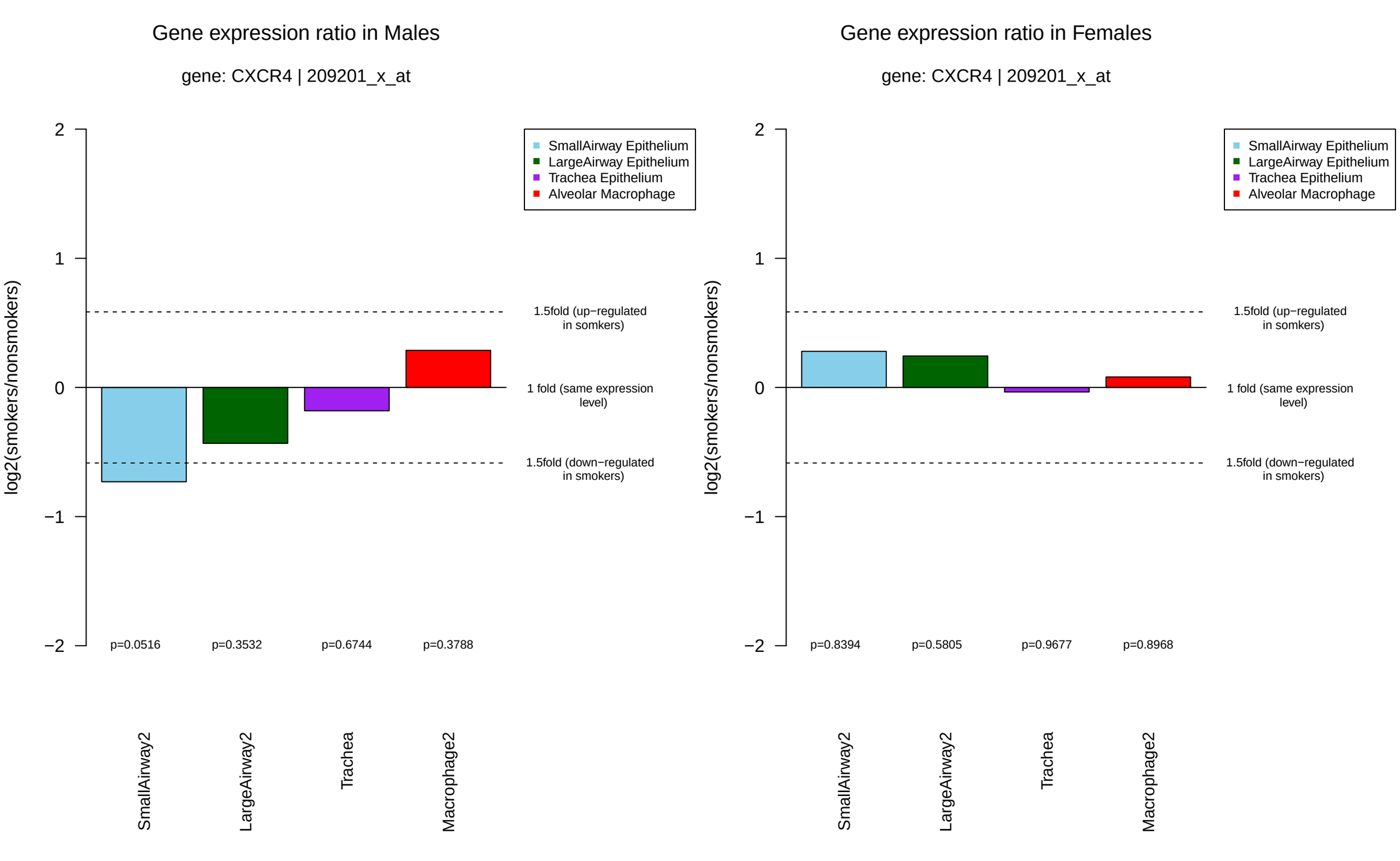
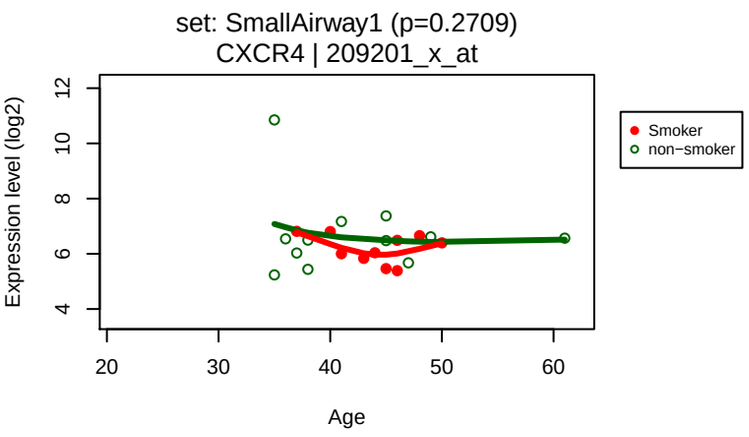
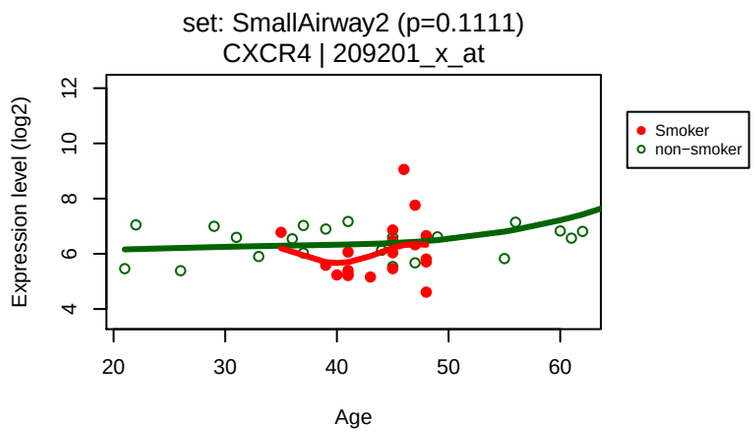
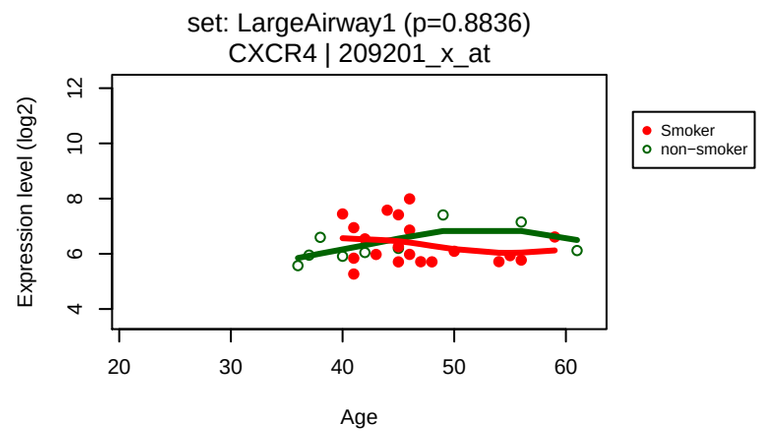

Gene expression of smokers and nonsmokers
gene: CXCR4 | 209201_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2709
p=0.1111
p=0.0501
p=0.8836
p=0.5334
p=0.7464
p=0.2781
p=0.4135
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

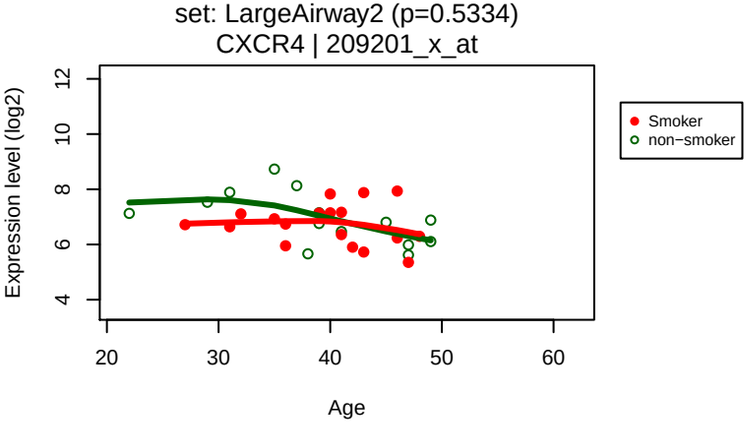

Gene expression ratio (smokers/nonsmokers)
gene: CXCR4 | 209201_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2709
p=0.1111
p=0.0501
p=0.8836
p=0.5334
p=0.7464
p=0.2781
p=0.4135
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

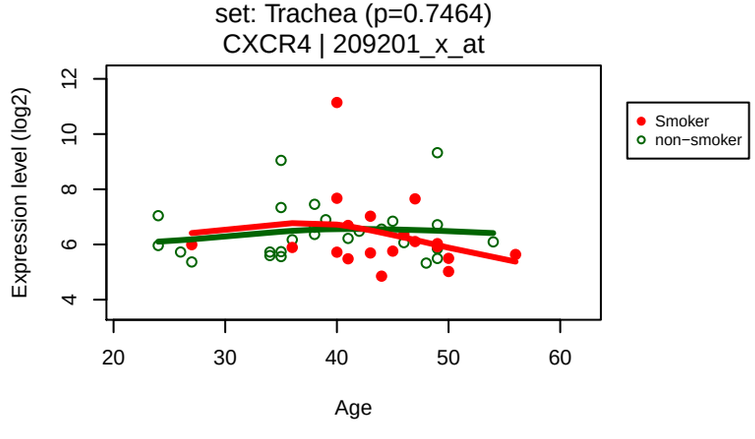

Gene expression ratio in Males
gene: CXCR4 | 209201_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0516
p=0.3532
p=0.6744
p=0.3788
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CXCR4 | 209201_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8394
p=0.5805
p=0.9677
p=0.8968
SmallAirway2
LargeAirway2
Trachea
Macrophage2

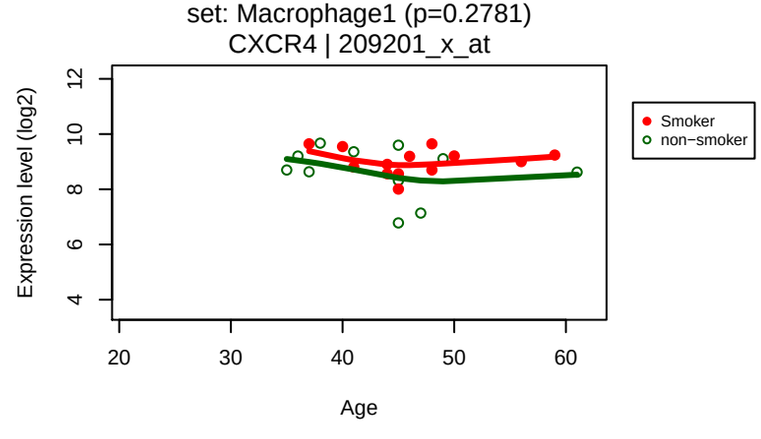

set: SmallAirway1 (p=0.2709)
CXCR4 | 209201_x_at
Expression level (log2)
Age
Smoker
non−smoker

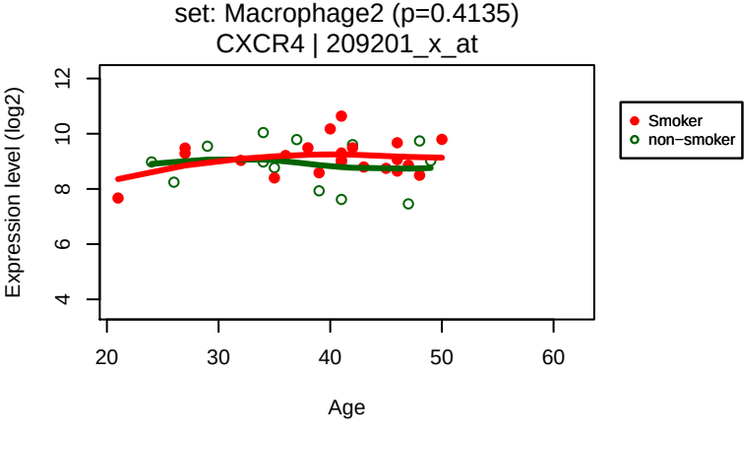

set: SmallAirway2 (p=0.1111)
CXCR4 | 209201_x_at
Expression level (log2)
Age
Smoker
non−smoker

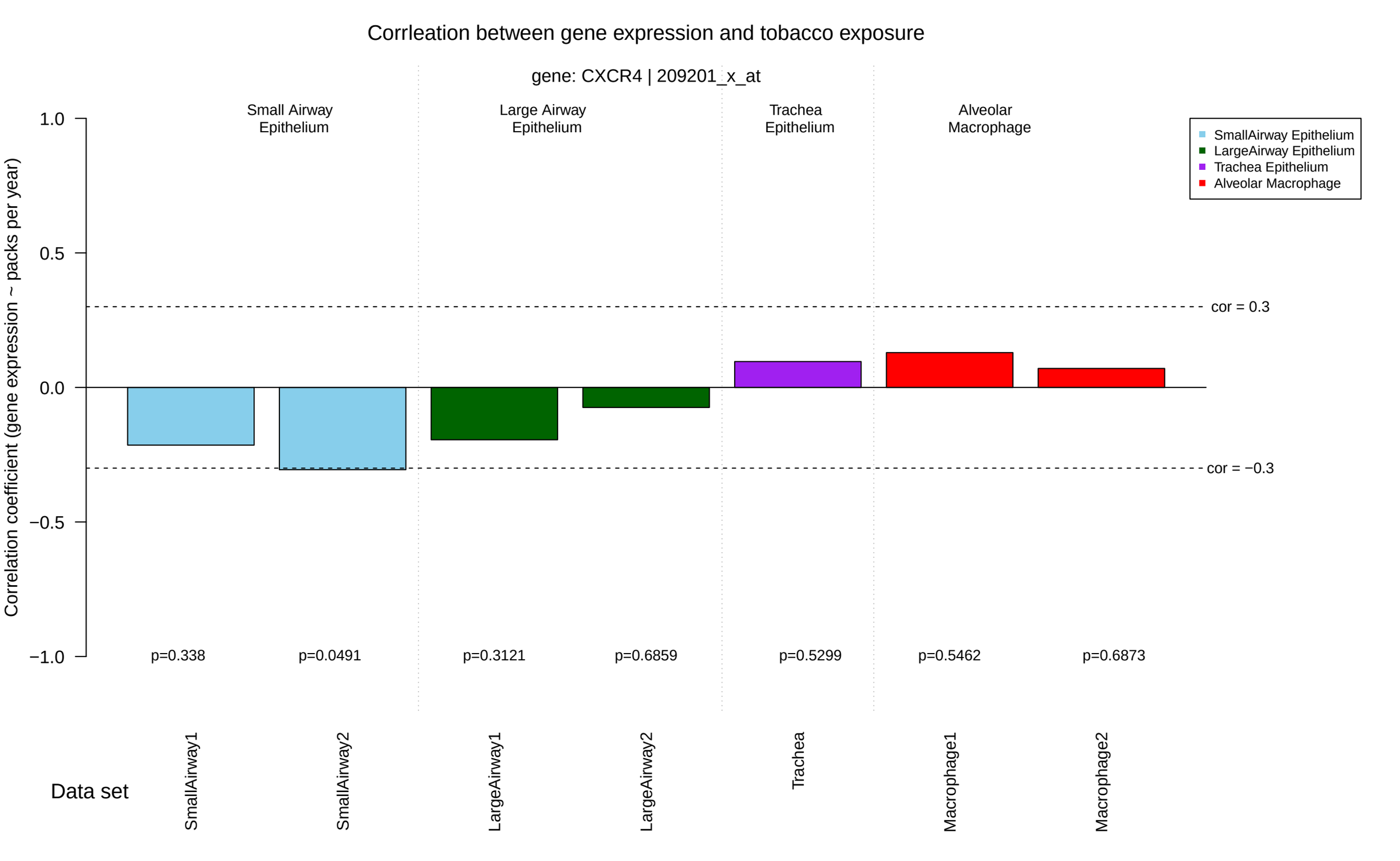

set: LargeAirway1 (p=0.8836)
CXCR4 | 209201_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5334)
CXCR4 | 209201_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7464)
CXCR4 | 209201_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2781)
CXCR4 | 209201_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4135)
CXCR4 | 209201_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CXCR4 | 209201_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.338
p=0.0491
p=0.3121
p=0.6859
p=0.5299
p=0.5462
p=0.6873
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2